

Lab 1: First-Order (RL/RC) Circuits

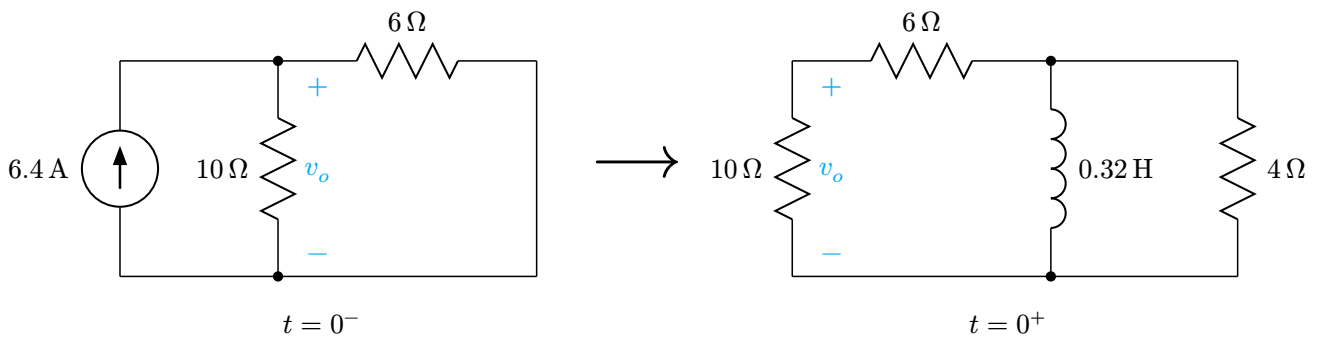
ECE2200L Section 04, Fall 2026

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No. 1

Step 1. Find $v_o(t)$ for $t \geq 0^+$



for $t = 0^-$:

$$10 \parallel 6 = 3.75\ \Omega$$
$$v_o(0^-) = 6.4(3.75) = 24\ \text{V}$$
$$i_L(0^-) = \frac{24}{6} = 4\ \text{A}$$

for $t > 0$:

$$i_L(0^+) = i_L(0^-) = 4\ \text{A}$$
$$R_{\text{eq}} = 4 \parallel (10 + 6) = 3.2\ \Omega$$
$$\tau = \frac{L}{R_{\text{eq}}} = \frac{0.32}{3.2} = 0.1\ \text{s}$$

Note that $i_L(\infty) = 0$ for natural response

$$i_L(t) = \cancel{i_L(\infty)}^0 + [i_L(0^+) - \cancel{i_L(\infty)}^0] e^{-t/\tau}$$

$$i_L(t) = i_L(0^+) e^{-t/\tau}$$

$$\boxed{i_L(t) = 4e^{-10t}\ \text{A}} \text{ for } t \geq 0$$

$$i_o(t) = \frac{4}{16 + 4} i_L(t) = 0.8e^{-10t}$$

$$-v_o(t) = 10i_o(t)$$

$$\boxed{v_o(t) = -8e^{-10t}\ \text{V}} \text{ for } t \geq 0$$

Step 2. Simulate the circuit and confirm the result in Step 1

[View simulation code](#)

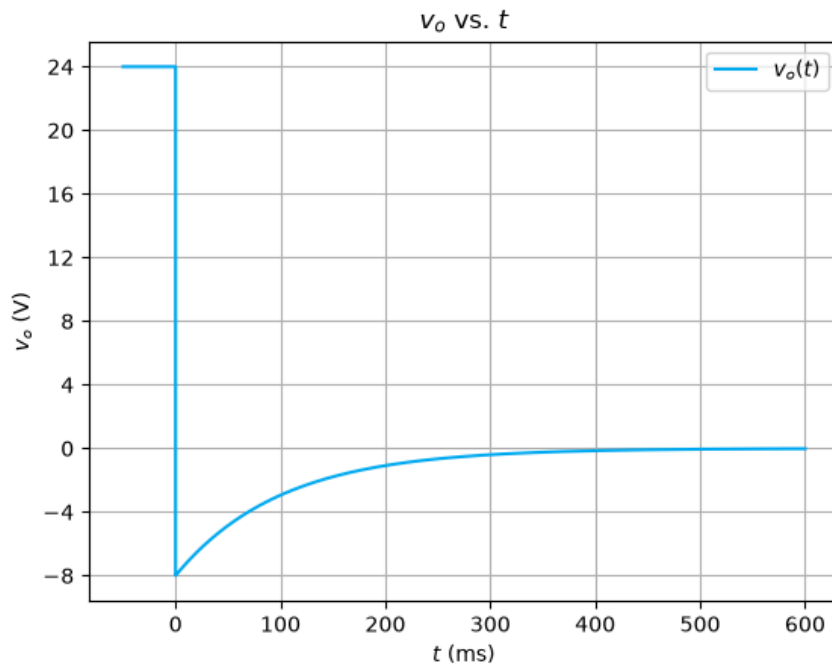


Figure 1: The simulation agrees with the theoretical first-order RL response. At $t = 0^+$, the inductor current remains continuous at 4A, while v_o changes instantaneously to -8V because resistor voltage is not required to be continuous. The response then decays exponentially toward 0V with a time constant of 0.1s. After approximately five time constants, or 0.5s, the response is essentially at steady state.

Step 3. Find the time at which 75% of the initial stored energy has been dissipated.

$$w(t) = \frac{1}{2}Li^2(t)$$

$$w(t) = 2.56e^{-20t} \text{ J}$$

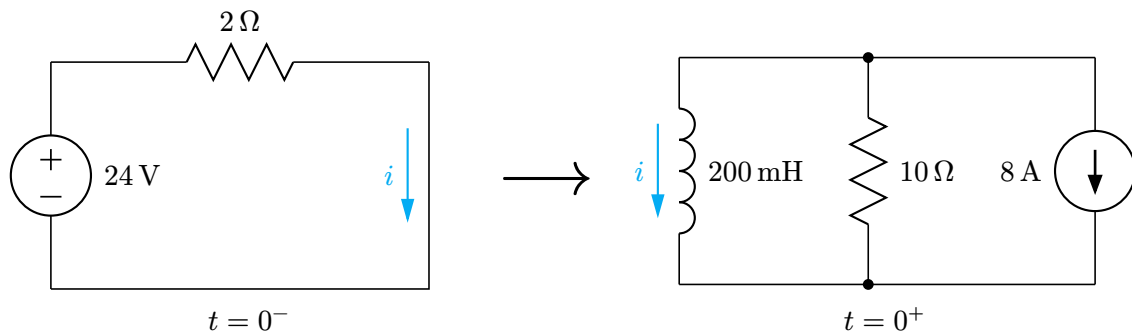
$$(1 - 0.75)w(0) = 2.56e^{-20t}$$

$$\Rightarrow t = -\frac{1}{20} \ln\left(\frac{0.25w(0)}{2.56}\right)$$

$$\boxed{t = 0.069 \text{ s}}$$

No. 2

Step 1. Find $i(t)$ for $t \geq 0^+$



for $t = 0^-$:

$$i(0^-) = \frac{24}{2} = 12\text{ A}$$

for $t > 0$:

$$i(0^+) = i(0^-) = 12\text{ A}$$

$$R_{\text{eq}} = 10\ \Omega$$

$$\tau = \frac{L}{R_{\text{eq}}} = \frac{0.2}{10} = 0.02\text{ s}$$

As $t \rightarrow \infty$, the inductor shorts and all 8 A from the current source flows through it. Since i is defined in the opposite direction as the source, we flip the sign.

$$i(\infty) = -8\text{ A}$$

$$i(t) = i(\infty) + [i(0^+) - i(\infty)]e^{-t/\tau}$$

$$i(t) = -8 + [12 - (-8)]e^{-t/0.02}$$

$$\boxed{i(t) = -8 + 20e^{-50t}\text{ A}} \text{ for } t \geq 0$$

Step 2. Simulate the circuit and confirm the result in Step 1

[View simulation code](#)

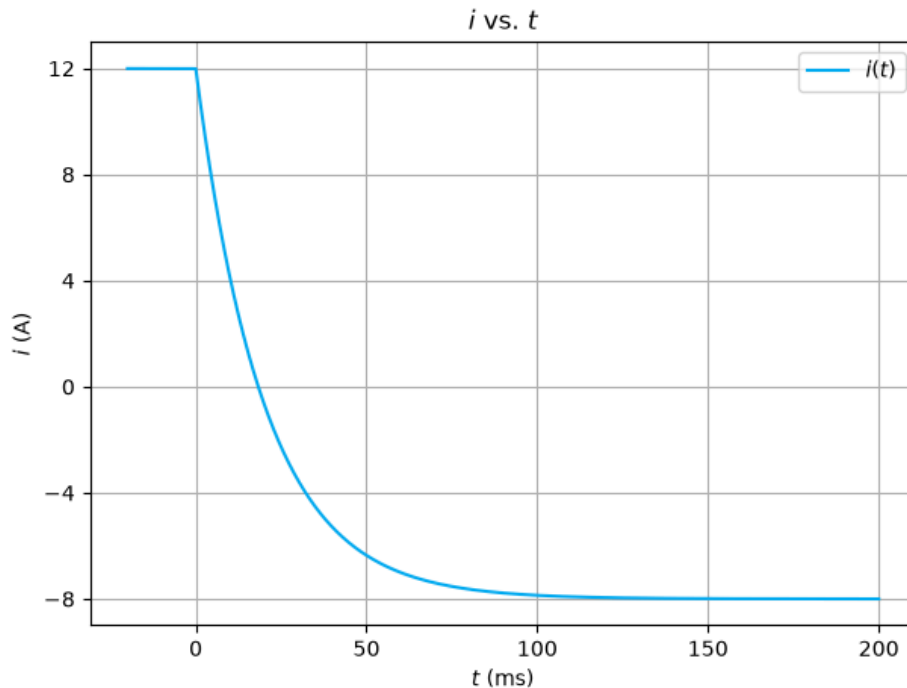
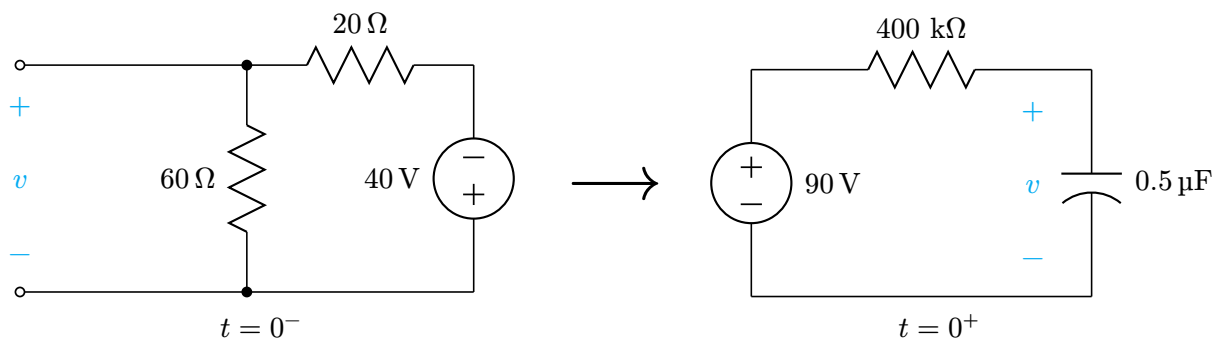


Figure 2: The simulated response begins at $i(0^+) = 12\text{ A}$, confirming continuity of inductor current. It then decreases exponentially toward the steady-state value of -8 A with $\tau = 20\text{ ms}$. The current crosses zero at approximately 18.3 ms ; the negative final value indicates that the steady-state current flows opposite to the reference direction chosen for i . By about $5\tau = 100\text{ ms}$, the circuit is essentially at steady state.

No. 3

Step 1. Find $v(t)$ for $t \geq 0^+$



for $t = 0^-$:

$$\begin{aligned} i &= \frac{40}{60 + 20} = 0.5\text{ A} \\ -v(0^-) &= 60i \\ \Rightarrow v(0^-) &= -30\text{ V} \end{aligned}$$

for $t > 0^+$:

$$v(0^+) = v(0^-) = -30 \text{ V}$$

$$R_{\text{eq}} = 400 \text{ k}\Omega$$

$$\tau = R_{\text{eq}}C = (400 \text{ k}\Omega)(0.5 \text{ }\mu\text{F}) = 0.2 \text{ s}$$

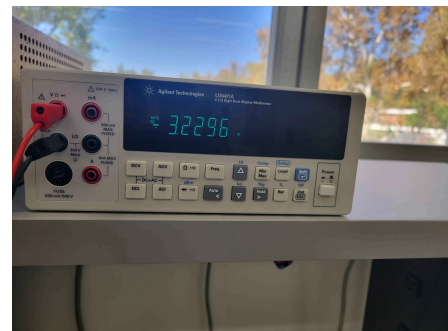
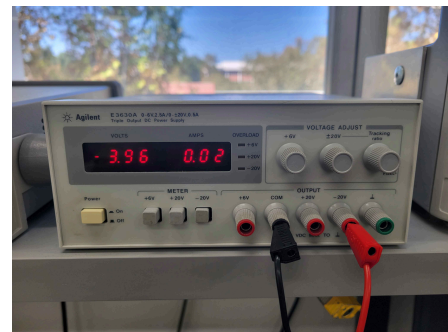
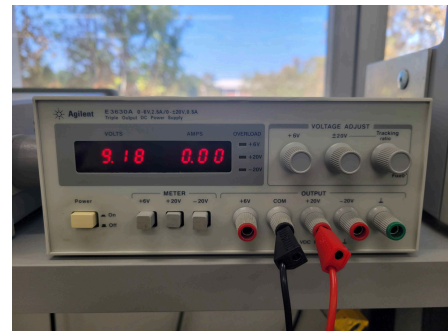
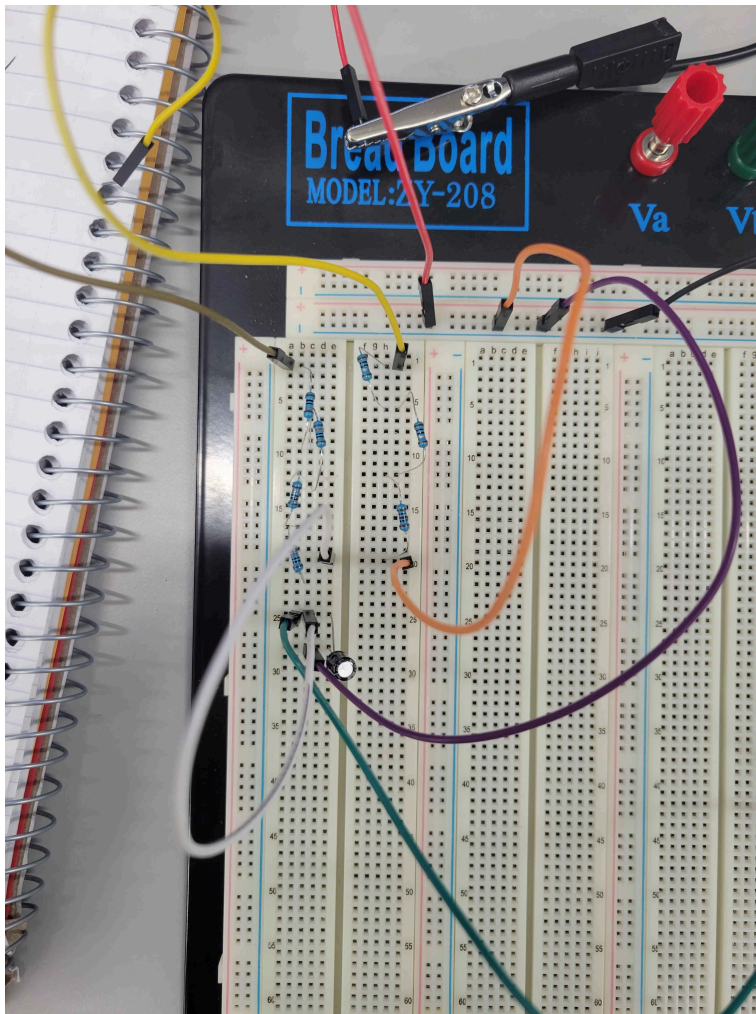
$$v(\infty) = 90 \text{ V}$$

$$v(t) = v(\infty) + [v(0^+) - v(\infty)]e^{-t/\tau}$$

$$v(t) = 90 + [-30 - 90]e^{-t/0.2}$$

$$\boxed{v(t) = 90 - 120e^{-5t} \text{ V}} \text{ for } t \geq 0$$

Step 2. Implement the circuit on a breadboard and confirm the result in Step 1



For the physical implementation, the source voltages were scaled down by a factor of 10 from the theoretical circuit. Therefore, the 40 V and 90 V sources were implemented as approximately 4 V and 9 V, respectively. Since the resistor and capacitor values were unchanged, the time constant remained

$$\tau = R_{\text{eq}}C = (400 \text{ k}\Omega)(0.5 \text{ }\mu\text{F}) = 0.2 \text{ s.}$$

Scaling down the source voltages by a factor of 10 also scales the capacitor voltage by the same factor. Thus, the experimental response was expected to be

$$v_{\text{exp}}(t) = 9 - 12e^{-5t} \text{ V}$$

with an initial voltage of approximately -3 V and a final steady-state voltage of approximately 9 V . The measured power-supply values of approximately -3.23 V (or -3.96 V) and 9.18 V are consistent with the scaled voltage sources used in the experiment. The experimental circuit therefore preserves the same first-order RC behavior and time constant as the original theoretical circuit, while operating at lower voltage levels.

Step 3. Simulate the circuit and confirm the result in Steps 1-2.

[View simulation code](#)

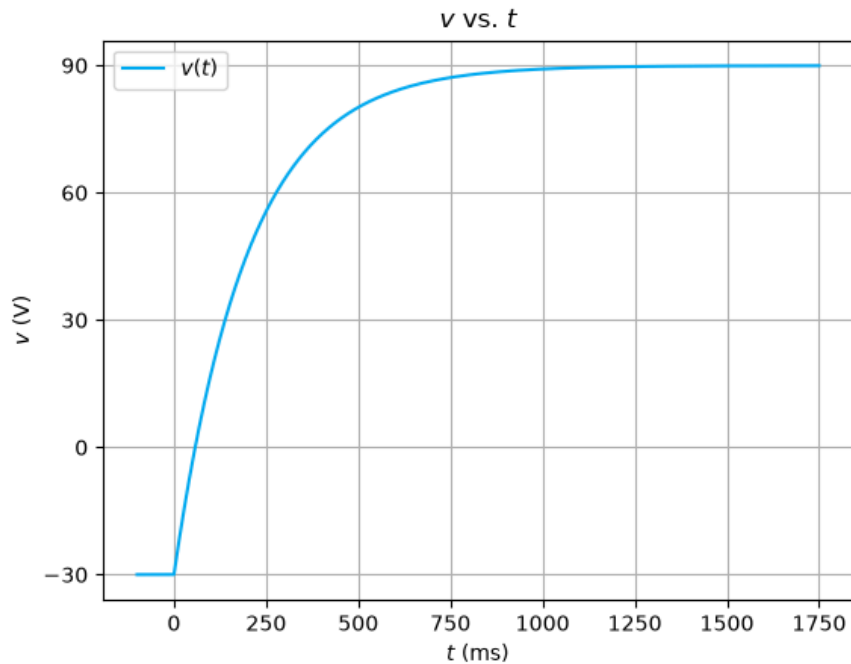


Figure 3: The capacitor voltage is continuous through switching, so $v(0^+) = v(0^-) = -30\text{ V}$. After switching, the capacitor charges toward the 90 V steady-state value through the $400\text{ k}\Omega$ resistance. The resulting time constant is $\tau = 0.2\text{ s}$. The theoretical response therefore rises exponentially from -30 V toward 90 V and is essentially settled after approximately $5\tau = 1\text{ s}$. The simulated waveform reflects this expected behavior.